Answer **all** the questions in the spaces provided.

1 The Earth may be considered as a uniform sphere of radius 6.37 x 10^3 km.

Cambridge is at a point on the Earth's surface that has a latitude of 52.2° north of the Equator, as shown in Figure 1.1.

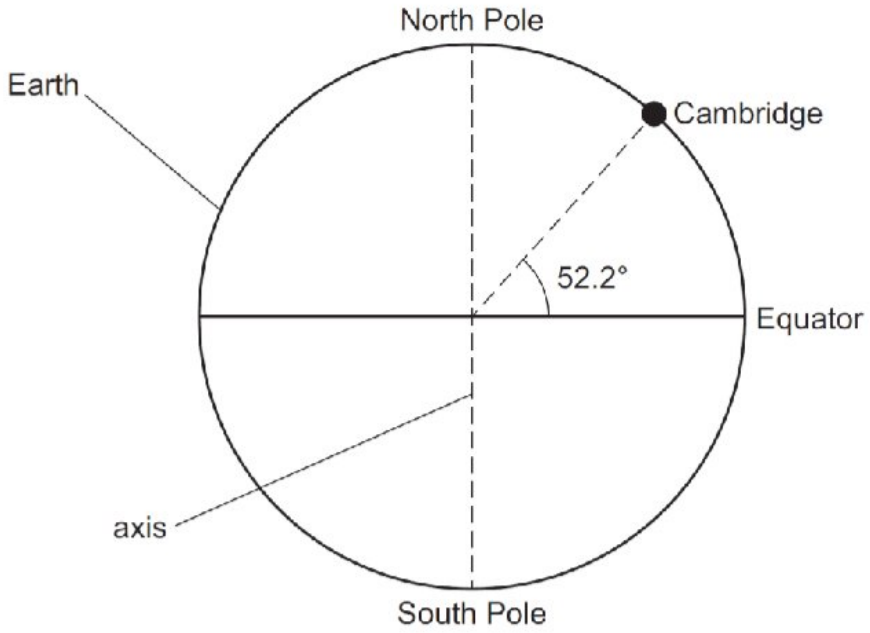


Figure 1.1

As the Earth spins on its axis, Cambridge moves in a circle that is parallel to the Equator but with a smaller radius.

(a) A student of mass 58.6 kg stands on horizontal ground in Cambridge.

(i) Determine the magnitude of the resultant force that acts to cause the circular motion of the student.

resultant force = ... N [3]

(ii) On Figure 1.2, draw labelled arrows from the student to show the directions of the forces that act on the student to cause the resultant force in **1(a)(i)**.

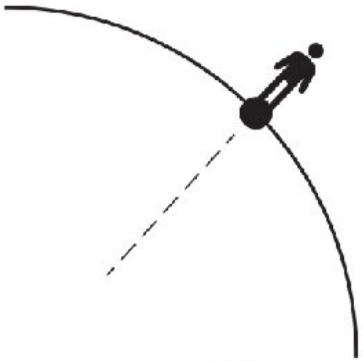


Figure 1.2

[2]

(b) Singapore has a small latitude north of Equator and can be assumed to be situated at the Equator. The Earth can be assumed to be a perfect sphere with uniform density.

State and explain how the apparent weight of another student of the same mass in Singapore compares to the student at Cambridge.

……..

……..

…….. [2]

[Total: 7]